CS683 Programming Assignment 1 — Task 1
2D Convolution: The Dhurandhar Way

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1 System Configuration

- **CPU:** Intel Core Ultra 5 125U (12 cores / 14 threads, 2P+8E+2LPE, max 4.3 GHz)
- **L1-D Cache:** 352 KiB total (10 instances), 64-byte lines, 8-way set associative
- **L2 Cache:** 10 MiB (5 instances)
- **L3 Cache:** 12 MiB (1 instance)
- **Compiler flags:** `-std=c++17 -O2 -fno-tree-vectorize -mavx2 -mfma`
- **OS:** Ubuntu Linux, kernel 7.0.0-30-generic (x86_64)
- **Pinning:** `taskset -c 0` (pinned to P-core 0 for consistent results on hybrid CPU)

2 Task 1A: Loop Unrolling and Reordering (1 point)

2.1 Motivation

[TODO: Explain what motivated the loop reordering and unrolling optimizations. Consider: reducing loop overhead, improving ILP, enabling out-of-order execution, improving spatial/temporal locality via reordering.]

2.2 Implementation Details

2.2.1 Loop Reordering

The naive convolution uses loop order `oy, ox, ky, kx` (output-major). We implemented a reordered version in `conv_reorder.cpp` with loop order `ky, kx, oy, ox` (kernel-major).

Key changes from naive:

- The output array must be zeroed before the computation begins, since each kernel position now *accumulates* into the output rather than computing the full sum for each output pixel.
- The kernel weight `ker[ky*K + kx]` is loaded once and reused across all $H \times W$ output pixels, improving temporal locality on the kernel.
- For each kernel position `(ky, kx)`, the inner loops stream sequentially over both the input and output arrays row by row, which is cache-friendly for both reads and writes.

2.2.2 Loop Unrolling (Naive Order)

We implemented four unrolling strategies in `conv_unroll.cpp` (strategies 1–4), all using the naive loop order `oy, ox, ky, kx`. All use an unroll factor of 4.

1. **Unroll ox (output columns) by 4:** Computes 4 adjacent output pixels simultaneously using 4 independent accumulators (`acc0`–`acc3`). Each accumulator sums the full $K \times K$ kernel independently, so the output is written once (not read-modify-written). This exposes instruction-level parallelism (ILP) since the 4 accumulations are independent. A scalar tail loop handles remaining columns when W is not a multiple of 4.

2. **Unroll kx (kernel columns) by 4:** Processes 4 kernel columns per inner-loop iteration, reducing loop overhead. Each iteration performs 4 independent multiply-adds into a single accumulator. A scalar tail handles the remainder when K is not a multiple of 4.

3. **Unroll ky (kernel rows) by 4:** Processes 4 kernel rows per iteration of the `ky` loop. Each iteration loads from 4 different input rows with their respective kernel row weights, accumulating into a single output value. Reduces `ky` loop overhead and exposes cross-row ILP. Tail handles remaining rows.

4. **Unroll ox + oy (2D output block, 4×4):** Processes a 4×4 block of output pixels at once, using 16 independent accumulators (`a00`–`a33`). For each kernel position, 4 input rows are loaded and 16 multiply-adds are performed. This maximizes ILP and register utilization. Tail loops handle edge rows and columns.

2.2.3 Loop Unrolling (Reordered Order)

We also implemented four unrolling strategies combined with the reordered loop order `ky, kx, oy, ox` (strategies 5–8 in `conv_unroll.cpp`). All require zeroing the output array first.

1. **Reorder + unroll ox by 4 (strategy 5):** For each kernel position, the inner `ox` loop is unrolled by 4, performing 4 read-modify-write accumulations per iteration. Amortizes loop overhead and improves throughput on the output stream.

2. **Reorder + unroll oy by 4 (strategy 6):** For each kernel position, processes 4 output rows simultaneously. Loads from 4 input rows (`r0`–`r3`) and accumulates into 4 output rows (`o0`–`o3`). The kernel weight broadcast is amortized across 4 rows.

3. **Reorder + unroll ox and oy (4×4 block, strategy 7):** Combines strategies 5 and 6. For each kernel position, processes a 4×4 block of output pixels, performing 16 read-modify-write accumulations per iteration.

4. **Reorder + unroll kx by 4 (strategy 8):** Instead of iterating `kx` one at a time, processes 4 kernel columns per iteration. For each group of 4 kernel positions, 4 different weights are loaded and 4 multiply-adds are performed per output pixel, sharing the output read-modify-write across 4 kernel positions.

2.3 Results

2.3.1 Reordering Speedup Table

Table 1: Loop reordering speedups (K=3)

Strategy	512	1024	2048	4096	8192
Reorder only	–	1.00x	0.88x	0.78x	0.78x
Reorder + unroll ox	2.00x	1.29x	1.21x	0.77x	0.96x
Reorder + unroll oy	1.27x	1.31x	0.86x	0.79x	0.76x
Reorder + unroll ox+oy	2.00x	1.43x	0.90x	0.66x	0.91x
Reorder + unroll kx	1.75x	1.42x	1.48x	1.21x	0.86x

Table 2: Loop reordering speedups (K=5)

Strategy	512	1024	2048	4096	8192
Reorder only	1.71x	1.20x	1.07x	1.04x	1.02x
Reorder + unroll ox	1.61x	1.50x	1.30x	0.90x	1.08x
Reorder + unroll oy	1.60x	2.29x	1.18x	1.34x	1.10x
Reorder + unroll ox+oy	1.97x	3.10x	2.74x	1.14x	1.28x
Reorder + unroll kx	2.33x	4.33x	1.39x	2.03x	1.61x

2.3.2 Unrolling Speedup Table (Naive Order)

Table 3: Loop unrolling speedups (naive loop order, K=3)

Strategy	512	1024	2048	4096	8192
Unroll ox	14.00x	2.95x	4.23x	3.84x	1.90x
Unroll kx	0.56x	0.63x	1.39x	0.75x	0.57x
Unroll ky	0.67x	0.69x	0.87x	0.68x	0.68x
Unroll ox+oy	–	2.79x	2.60x	2.38x	2.33x

Table 4: Loop unrolling speedups (naive loop order, K=5)

Strategy	512	1024	2048	4096	8192
Unroll ox	3.29x	3.17x	3.02x	2.94x	2.39x
Unroll kx	1.54x	2.97x	1.26x	1.24x	1.40x
Unroll ky	1.25x	3.03x	1.31x	1.48x	1.35x
Unroll ox+oy	3.93x	2.27x	1.91x	2.69x	3.08x

2.4 Effectiveness Metric

[TODO: Which metric did you use to quantify effectiveness and why? (e.g. speedup = naive_time / optimized_time, GFlop/s). Justify the choice. Discuss how speedup varies with matrix size and kernel size.]

2.5 Analysis

[TODO: Discuss: Which unrolling strategy works best and why? Does reordering help? How does the benefit change with matrix size (cache effects)? Which combination gives the best result?]

3 Task 1B: Tiling (1 point)

3.1 Profiling Baseline 2D Convolution

3.1.1 L1-D Cache Size

The L1 Data cache on this system is 352 KiB total across 10 instances (P-cores and E-cores have separate L1-D caches). Each L1-D has 64-byte cache lines and is 8-way set associative. When pinned to a single P-core (core 0), the effective L1-D available is approximately 48 KiB.

3.1.2 Baseline (Naive) L1-D Cache MPKI

Table 5: Naive convolution L1-D cache MPKI

Matrix Size	K=3 MPKI	K=5 MPKI
512x512	0.42	0.26
1024x1024	0.37	0.25
2048x2048	0.41	0.27
4096x4096	0.50	0.25
8192x8192	0.73	0.24

3.2 Implementing Tiled 2D Convolution

[TODO: Describe your tiling approach. What dimension(s) did you tile? (e.g. horizontal strips of height `TILE_H`). Explain the algorithm: zero the output strip, then iterate over K×K kernel elements, accumulating into the strip.]

[TODO: Explain the rationale: why does tiling help? (keeping the output strip in L1 cache across K×K passes reduces capacity misses).]

3.3 Experimenting with Tile Sizes

Table 6: Tiled convolution: speedup and MPKI for K=3

	Tile Height					
Matrix Size	8	16	32	64	128	256
Speedup over naive						
512x512	0.96x	2.60x	1.44x	1.52x	1.58x	3.17x
1024x1024	1.20x	1.29x	1.36x	1.36x	1.46x	1.10x
2048x2048	1.01x	1.06x	1.10x	1.33x	0.98x	1.09x
4096x4096	1.12x	1.01x	1.22x	0.90x	1.08x	0.98x
8192x8192	1.08x	1.29x	1.03x	1.06x	1.09x	1.05x
L1-D MPKI						
512x512	0.57	1.50	1.44	1.44	1.37	1.41
1024x1024	1.51	1.60	1.51	1.50	1.45	1.65
2048x2048	2.00	1.95	2.00	1.89	1.95	2.12
4096x4096	2.18	1.99	2.01	2.13	2.25	2.30
8192x8192	2.29	2.21	2.20	2.35	2.36	2.40

Table 7: Tiled convolution: speedup and MPKI for K=5

	Tile Height					
Matrix Size	8	16	32	64	128	256
Speedup over naive						
512x512	2.03x	1.39x	1.56x	1.17x	1.36x	1.00x
1024x1024	1.75x	1.16x	1.38x	1.50x	1.45x	1.35x
2048x2048	1.19x	1.19x	1.34x	1.35x	1.34x	1.33x
4096x4096	1.06x	1.19x	1.34x	1.33x	1.31x	1.25x
8192x8192	1.26x	1.19x	1.32x	1.24x	1.25x	1.03x
L1-D MPKI						
512x512	0.36	1.56	1.39	1.40	1.35	1.20
1024x1024	1.52	1.71	1.51	1.47	1.43	1.53
2048x2048	2.09	2.24	2.08	2.00	2.00	2.23
4096x4096	2.32	2.26	2.09	2.13	2.30	2.35
8192x8192	2.28	2.19	2.18	2.37	2.42	2.29

3.3.1 Best Tile Size

[TODO: Identify which tile size gives the best performance on your hardware. Justify in terms of cache efficiency and execution time.]

3.4 Plots

3.4.1 L1-D MPKI vs. Matrix Size

[TODO: Insert plot: L1-D MPKI vs. Matrix Size for different tile sizes. Include naive as baseline. One plot per K value (or combined).]

Figure 1: L1-D MPKI vs. Matrix Size for different tile sizes

3.4.2 Speedup vs. Matrix Size

[TODO: Insert plot: Speedup vs. Matrix Size for different tile sizes.]

Figure 2: Speedup vs. Matrix Size for different tile sizes

3.5 Analysis Questions

3.5.1 Q1: Changes in L1-D MPKI from Naive to Tiled

[TODO: Report the changes in L1-D MPKI observed when moving from naive to tiled 2D convolution. Justify your observations — why does tiling reduce (or not reduce) cache misses? Relate to the working set fitting in L1.]

3.5.2 Q2: How L1-D MPKI Varies Across Matrix Sizes and Tile Sizes

[TODO: Explain your findings in terms of the cache hierarchy and working set sizes. For example: for small matrices, the entire output may fit in cache even without tiling; for large matrices, smaller tiles keep the strip in L1. Discuss the sweet spot.]

3.5.3 Q3: Speedup Analysis

[TODO: Did you achieve a speedup? If yes, quantify the improvement and identify the contributing factors (reduced cache misses, better spatial locality). If not, analyze the limiting factors (e.g. overhead of zeroing, extra loop control) and propose possible solutions.]

4 Task 1C: SIMD (2 points)

4.1 Baseline Profiling

4.1.1 Instruction Count for Naive Implementation

Table 8: Naive convolution instruction count

Matrix Size	K=3 Instructions	K=5 Instructions
512x512	291M	625M
1024x1024	1,153M	2,490M
2048x2048	4,602M	9,966M
4096x4096	18,460M	39,783M
8192x8192	73,601M	158,802M

4.2 SIMD Implementation

4.2.1 128-bit (SSE)

[TODO: Describe the 128-bit SSE implementation. Which intrinsics did you use? (e.g. `_mm_loadu_ps`, `_mm_set1_ps`, `_mm_fmadd_ps`, `_mm_storeu_ps`). How many floats per vector? (4). Explain the vectorization strategy — which loop is vectorized (ox) and how.]

4.2.2 256-bit (AVX2)

[TODO: Describe the 256-bit AVX2 implementation. Which intrinsics? (`_mm256_loadu_ps`, `_mm256_set1_ps`, `_mm256_fmadd_ps`, `_mm256_storeu_ps`). How many floats per vector? (8). Any differences from SSE version besides width?]

4.2.3 512-bit (AVX-512)

AVX-512 is not available on the Intel Core Ultra 5 125U. Running `lscpu | grep avx512` returns no results, confirming the absence of AVX-512 support. Therefore, only 128-bit (SSE) and 256-bit (AVX2) SIMD implementations were tested.

4.3 Instruction Count for SIMD Implementations

Table 9: Instruction counts: naive vs. SIMD (K=3)

Matrix Size	Naive	SSE 128-bit	AVX2 256-bit
512x512	291M	343M	320M
1024x1024	1,153M	1,366M	1,272M
2048x2048	4,602M	5,460M	5,067M
4096x4096	18,460M	21,918M	20,249M
8192x8192	73,601M	87,366M	81,077M

Table 10: Instruction counts: naive vs. SIMD (K=5)

Matrix Size	Naive	SSE 128-bit	AVX2 256-bit
512x512	625M	741M	687M
1024x1024	2,490M	2,955M	2,734M
2048x2048	9,966M	11,884M	10,907M
4096x4096	39,783M	47,522M	43,662M
8192x8192	158,802M	188,744M	174,304M

4.4 Speedup Results

Table 11: SIMD speedups over naive (K=3)

Matrix Size	SSE 128-bit	AVX2 256-bit
512x512	2.81x	17.00x
1024x1024	3.70x	9.37x
2048x2048	4.85x	6.12x
4096x4096	4.30x	5.89x
8192x8192	4.24x	8.28x

Table 12: SIMD speedups over naive (K=5)

Matrix Size	SSE 128-bit	AVX2 256-bit
512x512	2.25x	6.28x
1024x1024	3.97x	9.41x
2048x2048	3.66x	5.90x
4096x4096	4.29x	7.53x
8192x8192	3.82x	6.78x

4.5 Plots

[TODO: Insert plot: Speedup vs. Matrix Size for different SIMD widths. Choose matrix sizes that allow you to clearly observe trends.]

Figure 3: SIMD speedup vs. Matrix Size for different SIMD widths

4.6 Analysis Questions

4.6.1 Q1: Change in Instruction Count from Naive to SIMD

[TODO: Report the change in the number of instructions when moving from naive to the SIMD implementation. Justify your observations — do instructions decrease? By how much? Is the reduction proportional to the SIMD width (4x for SSE, 8x for AVX2)? If not, why? (loop overhead, scalar remainder, setup instructions).]

4.6.2 Q2: Speedup Analysis

[TODO: Did you achieve any speedup? If so, how much, and what contributed to it? (reduced instruction count, better throughput per cycle, FMA combining multiply+add). If not, what were the reasons? (memory bandwidth bottleneck, cache misses dominating).]

4.6.3 Q3: Choice of SIMD Intrinsics

[TODO: Which SIMD intrinsics did you use? Justify your choice of functions. For example:

- **`_mm256_loadu_ps`: unaligned load — needed because input offsets are not always 32-byte aligned.**
- **`_mm256_set1_ps`: broadcast kernel weight to all lanes.**
- **`_mm256_fmadd_ps`: fused multiply-add — combines multiply and add into a single instruction, reducing latency and instruction count.**
- **`_mm256_storeu_ps`: unaligned store of result.**

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5 Task 1D: Sabka Saath Sabka Vikaas (2 points)

5.1 Combined Optimized Implementation

[TODO: Describe your combined implementation in `conv_optimized.cpp`. Which techniques did you combine? (e.g. SIMD + tiling, SIMD + unrolling, SIMD + reordering, all of them). Explain how these techniques complement each other for synergistic gains — i.e. the combined benefit is greater than the sum of individual optimizations.]

5.2 Speedup Results

Table 13: Combined optimized speedup over naive

Matrix Size	K=3 Speedup	K=5 Speedup
512x512	2.27x	–
1024x1024	7.36x	8.21x
2048x2048	11.49x	10.92x
4096x4096	6.92x	8.49x
8192x8192	6.83x	7.56x

5.3 Final Performance Summary Plot

[TODO: Insert plot comparing the best performance achieved by each optimization technique across varying matrix sizes. The plot should include the following series (as in Figure 1.1 of the problem statement):

- **Loop unrolling (best variant)**
- **Loop reordering (best variant)**
- **Tiling (best tile size)**
- **SIMD (best width)**
- **Tiling + SIMD (combined)**

Y-axis: normalized speedup (no optimization = 1.0). X-axis: matrix sizes. Create separate plots for K=3 and K=5 (or one combined).]

Figure 4: Comparison of all optimization techniques — Speedup vs. Matrix Size

[TODO: You can try different combinations and expand the plot accordingly (e.g. reordering + unrolling, tiling + unrolling, etc.).]

5.4 Analysis

[TODO: Discuss how these techniques complement each other:

- **Why does combining techniques yield more than the sum of parts?**
- **Which combination gave the best speedup?**
- **What is the bottleneck of your best implementation? (compute-bound? memory-bound? Roofline analysis?)**
- **How does the gain/loss of each technique vary with matrix size and kernel size?**

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